

## Tossing a die

favourable outcomes  $\rightarrow$  If face of the die is an even number

$$E = \{2, 4, 6\}$$

All possible outcomes  $\rightarrow \{1, 2, 3, 4, 5, 6\}$

$$\frac{\# \text{ of favourable outcomes}}{\text{Total number of outcomes}} = \frac{3}{6} = \frac{1}{2}$$

Toss a biased coin 10 times

let the actual classical probabilities  
be as follows:

$$P(H) = \frac{1}{3}$$

$$P(T) = \frac{2}{3} = \underline{0.66.}$$

When we toss the coin 10 times, we may  
see 8 tails  $\left(\frac{8}{10} \neq \frac{2}{3}\right)$

When we toss the coin 100 times, we  
see 59 tails  $\left(\frac{59}{100} \neq \frac{2}{3}\right)$

When we toss it 1000 times we may see  
650 tails  $\left(\frac{650}{1000} \approx 0.66\right)$

$S = \text{sample space} = \text{set of all possible}$

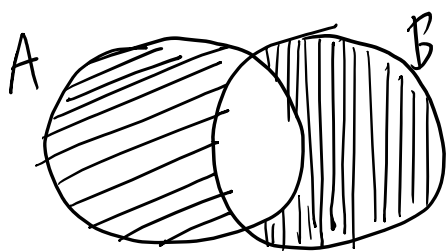
outcomes

$E = \text{a subset of } S$

Where does the addition rule come from?

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

If comes from the third axiom of  
the axiomatic approach to probability



Mutually exclusive  
 $\underline{A_{only} \cap A \cap B = \emptyset}$   
 (null set)

$$A \cup B = A_{only} \cup A \cap B \cup B_{only}$$

$A_{only}, A \cap B, B_{only}$  are mutually exclusive

$P(A \cup B)$  by the third axiom

$$= P(A_{only}) + P(A \cap B) + P(B_{only})$$

Since  $A_{only}, A \cap B$  and  $B_{only}$  are mutually exclusive events

The event A is  $A_{only} \cup A \cap B$

$$P(A) = P(A_{only}) + P(A \cap B) \text{ by the third}$$

axiom since  $A_{only}$  and  $A \cap B$  are mutually exclusive

$$P(A \cup B) = \underbrace{P(A_{\text{only}})}_{\downarrow} + \underbrace{P(B_{\text{only}})}_{\downarrow} + P(A \cap B)$$

$$P(A) - P(A \cap B) \quad P(B) - P(A \cap B)$$

$$P(A \cup B) = P(A) + P(B) - 2P(A \cap B) + P(A \cap B)$$

$$\boxed{P(A \cup B) = P(A) + P(B) - P(A \cap B)}$$

$$S = A \cup A^c$$

$$P(S) = 1 = P(A \cup A^c)$$

By the third axiom, since  $A$  and  $A^c$  are mutually exclusive

$$P(S) = P(A \cup A^c) = P(A) + P(A^c) = 1$$

$$\therefore \boxed{P(A) = 1 - P(A^c)}$$

There are 36 possible outcomes in the sample space of tossing two dice

$$\left. \begin{array}{l} (1,1) \quad (1,2) \quad \dots \quad (1,6) \\ (2,1) \quad (2,2) \quad \dots \quad (2,6) \\ \vdots \\ (6,1) \quad (6,2) \quad \dots \quad (6,6) \end{array} \right\}$$

We are assuming that each of these possible outcomes is equally likely,

$$P(\text{Outcome} = i) = \frac{1}{36}$$



$$P(X=2) = P\{(1,1)\} = \frac{1}{36}$$

$$\begin{aligned} P(X=3) &= P\{(1,2), (2,1)\} = P(1,2) \\ &\quad + P(2,1) \\ &= \frac{1}{36} + \frac{1}{36} \\ &= \frac{2}{36} \end{aligned}$$

$A$  = event denoting that a student  
passes JEE Advanced

$S$  = event denoting that a student  
got a high score in the 12th std  
school leaving examination

$$\frac{P(A) = 0.1 \quad [\text{prior}]}{P(A/S) > P(A) \text{ in general}} \\ \text{posterior}$$

cloudy & Sunny are mutually  
exclusive

$$P(\text{cloudy} | \text{sunny}) = 0$$

mutually  
exclusive  
events  
interpreted  
through  
conditional  
probability

$$P(\text{sunny} \cap \text{cloudy}) = 0$$

Since sunny & cloudy are  
mutually exclusive events

$$P(\text{sunny} \cap \text{cloudy}) = P(\text{sunny})P(\text{cloudy} | \text{sunny})$$

Assuming that  $P(\text{sunny}) \neq 0$ , we  
see that  $P(\text{cloudy} | \text{sunny}) = 0$

$$P(\overset{D}{\text{a person has a disease}}) = 0.2$$

[20% of randomly selected people are found to have the disease]

Let us now look at the class of people who have undergone a blood test B and which came out negative.

$$P(D/B = \text{false}) = \underline{0.05}$$

## Multiplication rule

$$P(A \cap B \cap C) = P(A) P(B/A) P(C/A \cap B)$$

$$P(A \cap \overbrace{B \cap C}^E) = P(A) P(\overbrace{B \cap C}^E / A) \quad \begin{array}{l} \text{using definition} \\ \text{conditional} \\ \text{probability} \end{array}$$

$$= P(A) \{ P(B/A) P(C/A \cap B) \}$$

$$P(A \cap E) = P(A) P(E/A)$$

$$P(E/A) = P(B \cap C/A) = P(B/A) P(C/\underbrace{A, B}_{A \cap B})$$

$$P(C/A, B) = \frac{P(B, C/A)}{P(B/A)}$$

PC

$$P(C/A \cap B) = \frac{P(A \cap B \cap C)}{P(A \cap B)}$$

$$P(E/F) = \frac{P(E \cap F)}{P(F)}$$

When  $E = C$  and  $F = A \cap B$  the 2nd equation becomes the first equation

$$P(E/F) = \frac{P(E \cap F)}{P(F)}$$

Let  $F = A \cap B$ .

$$P(E/A \cap B) = \frac{P(E \cap A \cap B)}{P(A \cap B)}$$

$$P(E/A \cap B) = \frac{P(E \cap A \cap B)}{P(A)P(B/A)}$$

$$P(E \cap A \cap B) = P(A)P(B/A)P(E/A \cap B)$$

Multiplication rule comes from repeated application of the definition of conditional probability